

SHRIDHAR PATHAK

shripathmit@gmail.com

www.linkedin.com/in/shridhar-pathak-19277927

PROFESSIONAL SUMMARY

Senior AI Engineering Leader with 15+ years building and scaling distributed systems and AI platforms across consumer tech and regulated enterprise environments. Currently Sr. Director of AI at Moderna and former senior AI leader at Amazon Alexa, responsible for multi-agent conversational AI systems operating at global scale.

Proven record defining GenAI strategy, building conversational agent frameworks, launching AI-powered products across 10+ countries, and leading engineering organizations delivering high-reliability, low-latency AI infrastructure. Executive MBA from MIT Sloan, combining deep technical leadership with enterprise-scale transformation strategy.

Expertise spans LLM orchestration, multi-agent systems, developer platforms, MLOps, AI governance, cloud-native AI infrastructure, and enterprise AI adoption.

CORE LEADERSHIP & TECHNICAL EXPERTISE

Conversational Agents & Multi-Agent Systems, LLM Orchestration, Tool Calling, Fine-Tuning Strategy, Applied Generative AI Infrastructure, AI Platform Architecture & Developer Experience, Cloud-Native ML Systems (AWS, Distributed Microservices), AI Governance & Enterprise Compliance, Engineering Org Design & Global Delivery, Cross-Org Strategy (Research, Product, Infra, Legal, External Partners), MLOps & Production ML Lifecycle.

PROFESSIONAL EXPERIENCE

Senior Director, AI & ML: Moderna | Cambridge, MA | 2023 – Present

- Leading AI strategy and engineering for enterprise GenAI and multi-agent systems: Defined and executed Moderna's generative AI roadmap for 20+ enterprise applications, including conversational agents, function calling, and LLM orchestration workflows for R&D, manufacturing, and clinical operations (15+ years equivalent experience through EMBA-accelerated business strategy.)
- Built and led cross-functional AI/ML engineering teams (35+ engineers) delivering production LLM-based applications integrated with cloud infrastructure (AWS, Azure) using Python and MLOps pipelines (30+). Architected multi-agent conversational systems leveraging fine-tuning, distillation, and RAG (retrieval-augmented generation) to automate knowledge work across regulated enterprise workflows for all document systems at Moderna.
- Exec MBA Impact: Partnered with C-suite and business unit leaders to translate AI technical capabilities into measurable business outcomes, accelerating go-to-market strategy and stakeholder buy-in across pharmaceutical R&D, quality, and regulatory teams. Established strategic partnerships with OpenAI, AWS, and cloud AI vendors to integrate cutting-edge generative AI technologies into enterprise platforms.

Amazon: Senior Manager & Tech Lead, Alexa AI and Multimodal Experiences | Seattle, WA | 2018 – 23

Senior engineering leader responsible for large-scale conversational AI and multi-agent orchestration frameworks powering Alexa globally, managing 35+ direct report and 110+ partner team reports spread across 8+ countries. Key Leadership Contributions:

- Multi-Agent Conversational Framework (Foundational Work for Agentic Systems): Architected and launched Alexa's first multi-agent conversational switching framework. Enabled seamless orchestration across celebrity personalities (Melissa McCarthy, Shaquille O'Neal, Amitabh Bachchan) and partner agents (Disney). Designed contextual aggregation and routing services handling millions of low-latency voice requests daily. Built scalable architecture supporting dynamic agent selection and personality switching.
- Global AI Product Expansion: Launched Alexa Voice Shopping across US, India, Japan, Canada, France, Italy, Spain, Mexico, Ireland, and Australia. Delivered international AI product rollouts requiring localization, regulatory adaptation, and infrastructure scaling.
- Applied ML Systems Led integration of speech recognition, NLP, contextual understanding, and response orchestration. Defined APIs and service endpoints enabling ML-powered interactions in the Amazon India retail app (millions of DAU). Balanced personalization, privacy, latency, and reliability at global scale.
- Engineering Leadership Built and scaled multi-disciplinary teams (35+ engineers across regions). Drove cross-org alignment across Alexa AI, retail, device, and international engineering teams. Set architectural standards for reliability and developer extensibility. Impact: Delivered production conversational AI systems used by millions of customers globally, forming the backbone of agentic consumer AI experiences.

Amazon: Software Development Engineer: 2012 – 2016

Built distributed services for Amazon Register (POS platform). Designed onboarding, payments, and integration systems for merchants. Early career foundation in high-scale, fault-tolerant distributed systems.

Tata Institute of Fundamental Research (TIFR) / CERN GRAPES-3 | Junior Research Fellow|2009 – 2010

Designed detector-monitoring algorithms under CERN experimental frameworks. Built scientific computing systems using C++ and ROOT.

Noesis: Enterprise Agent Workforce Platform | Tech Advisor-2025 – Present

Conceived and architected a workforce management layer for enterprise-scale AI agents, positioning AI systems as a managed "digital workforce." Designed Policy-as-Code and runtime supervision frameworks enabling real-time monitoring, intervention, and compliance for LLM-driven multi-agent systems. Built architectural blueprint for scalable agent orchestration including tool use, function calling, memory, and workflow chaining. Defined performance benchmarking models to measure agent quality, reasoning efficiency, and enterprise ROI. Positioned platform strategy for regulated industries requiring transparency, auditability, and safe GenAI adoption.

EDUCATION

MIT Sloan School of Management — Executive MBA (2026)

Focus: System Dynamics, Competitive Strategy, Platform Economics, Digital Transformation, and AI-Driven Market Strategy.

University of Texas at Dallas — Master of Science, Computer Science (2012)

Pune Institute of Computer Technology — Bachelor of Engineering, Computer Science (2009)

AWARDS & LEADERSHIP RECOGNITION

Stanford Seed Consultant — Stanford Graduate School of Business

Selected strategic advisor within the Seed Transformation Network, partnering with high-growth global founders to scale operations, strengthen AI-enabled business models, and drive sustainable expansion.

Engineering Leadership Development — Rice University

Advanced executive training in leading high-performing technical teams, organizational alignment, and cross-functional relationship management.

Kishore Vaigyanik Protsahan Yojana (KVPY) Fellow — Government of India

Nationally recognized Young Scientist Scholar for academic and research excellence in engineering.